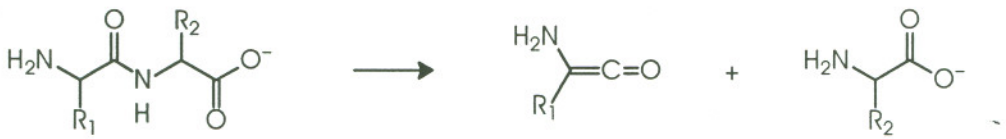
Organic Chemistry
Cumulative Examination
September 17, 1994

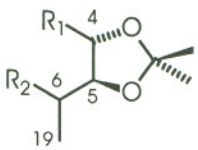
1. Our most recent departmental seminar featured Professor Ralph Becker, who talked about the photochemical processes involved in vision. One of the key step is the isomerization of the *cis* double bond in 11-*cis*-retinal to the trans conformation. As you well know, rotation around carbon–carbon double bonds is not an easy process. Using the formalism of **molecular orbital theory**, describe how this process can be achieved in the context of Professor Becker's talk. (Just describe the general process; you don't need to use retinal in your explanation.)

2. *J. Am. Chem. Soc.* **1994**, *116*, 7787. This paper describes mass-spectrometry studies of the dissociation of small deprotonated peptides in the gas phase. For peptides with non-acidic side chains, the authors proposed the following reaction:



Describe the steps involved in this reaction using the usual curved-arrow/electron-pushing diagrams. Remember this is a *gas-phase* reaction, so invoking such things as solvent participation is not allowed.

3. *J. Am. Chem. Soc.* **1994**, *116*, 7631. This paper describes structure determination of microsclerodermins, which are natural products isolated from a deep water sponge. During the course of the work, some derivatives were prepared. The picture below illustrate a fragment from one of these derivatives. The R groups are actually large, complicated structures. The numbering of the atoms is from the numbering in the original compound.



What is the stereochemical designation at C-4 and C-5, R or S? (For the purpose of prioritizing substituents, you may treat the R's as if they were ethyl groups.)

Note that I have **not** indicated the stereochemistry at C-6. That particular stereochemistry was determined from NOE data. The relevant correlations are betwen the protons at C-4 and C-6, C-5 and C-6, C-5 and C-19. So what is the stereochemistry at C-6, R or S? Show your reasoning.